

Context-Aware Selection of AI-generated Audio Descriptions for Blind and Low Vision Users

Melanie Medina

Data Science Major

melaniemedina2028@u.northwestern.edu

Faculty mentor: Shreeya Behera

Department of Statistics and Data Science, Assistant Professor of Instruction

shreeya.behera@northwestern.edu

Introduction: Audio description (AD), also known as video description, is a way of making videos accessible for blind and low vision users. It involves narrating visual events in a scene, such as setting, characters, actions, scene changes, and other stimuli. In films and television shows, producing AD requires significant effort from experts such as AD writers, voice actors, producers, and audio engineers. However, the majority of independent films and user-created internet content does not provide AD, which widens the accessibility gap for blind and low vision users (Li et al., 2025). Even with volunteer platforms such as YouDescribe, human description alone is insufficient for covering the massive amount of inaccessible online content. AI-generated AD can help fill this gap. In recent years, advances in natural language processing and vision-language models have enabled the rapid development of AD generation (Gao et al., 2024). AD generation is facilitated by multimodal large language models (MLLMs), which are AI systems that can interpret visual content and produce natural language descriptions. New AD generation models demonstrate significant description capabilities, but still struggle with avoiding overlap with dialogue, reducing redundancy, and fitting descriptions into narrow silence windows (Ye et al., 2024). This project builds on those developments by focusing on the selection problem rather than the generation problem. Instead of generating new AD from scratch, I will use existing AD models and apply a context-aware selection layer that determines which candidate descriptions should be inserted into the final audio track.

Background: Audio descriptions are generally more difficult to generate than auto-captions (Kala et al., 2025). Unlike captions, AD must be timed around dialogue, sound effects, and scene pacing, which makes the task more sensitive to context. A description can be grammatically correct and still fail as AD if it repeats spoken dialogue, covers information the listener can already infer, or is inserted when there is not enough silence (Li et al., 2025).

New AD generation models attempt to address these problems via different mechanisms. For example, the 2025 VideoA11y model combines natural language processing (NLP) with accessibility guidelines to generate AD that is better oriented toward blind and low vision users (Li et al., 2025). Another 2025 model, DistinctAD, focuses on reducing redundancy by taking neighboring dialogue into account (Fang et al., 2025). More broadly, a recent review of AD generation finds that newer models often follow a three-step pattern: dense video captioning, AD post-editing based on guidelines, and automatic quality checking, but notes that it remains challenging to use these AD outputs in actual viewing settings (Gao et al., 2024).

Together, these publications suggest that there are now multiple ways to generate AD, but less is known about how to select descriptions that fit in context. Existing work often evaluates generation quality using quantitative metrics, but there is little focus on lightweight methods that use contextual information to select outputs from existing generators (Kala et al., 2025). My project targets this selection layer. Designing a selector model with relevant contextual features may help improve accessibility without having to retrain large models from scratch.

Therefore, my research question is: **what kind of selector model and which contextual features are most effective for selecting AI-generated audio descriptions among a set of candidates for blind and low vision users?**

Methodology: To address my research question, I will build a processing pipeline with three phases: Phase A, Phase B, and Phase C.

Phase A is generation and feature extraction. The purpose of this phase is to produce a dataset where each row corresponds to a scene + candidate AD pair (see Table B1 in Appendix B). I will use a pretrained AD generation model, VideoA11y, to produce three candidate descriptions per video segment, since it was explicitly designed for accessible video description and already incorporates AD guidelines and multimodal context (Li et al., 2025). The three description variations will come from different prompting styles: one prompt will ask for an objective description, one will ask for description that follows AD guidelines (W3C, n.d.), and one will provide AD guidelines as well as an automatic speech recognition (ASR) transcript of

the dialogue. ASR converts spoken audio into text with time-aligned timestamps, which I will use to create timing-based features such as speech duration and silence duration. Finally, I will use a pretrained vision-language model (VLM), such as a Vid2Seq-style model pretrained on video data (Yang et al., 2023), to count how many major visual events happen in a scene. All other contextual features will be calculated from these basic elements.

Phase B is selection, and it will be the main contribution of the project. I will train a lightweight machine learning model that takes, as input, the features from Phase A for each candidate AD and predicts a quality score for that candidate. For each scene, the selector model will pick the highest-scoring description for the final audio track. I plan to experiment with several model architectures: linear regression, which learns a weighted combination of features to predict an output; random forests, which combine many simple decision rules to capture non-linear interactions between features; and gradient-boosting, which builds a sequence of small models where each one fixes errors from the previous one.

Phase C is rendering. This layer will take the selected descriptions and place them into the audio timeline so that the final AD track spans the length of each video and can be listened to as a complete audio experience.

My training and evaluation data for the selection model will be derived from existing human AD datasets. I plan to start with QuerYD (Oncescu et al., 2020), since it is designed for machine learning with temporal information, and YuWA, which contains a large collection of volunteer-produced AD transcriptions on YouTube videos (Pitcher-Cooper et al., 2024).

Evaluation will follow two stages: statistical and experimental. First, I will conduct a statistical evaluation of the selection model. The primary benchmark for success will be how often the selected candidate (the one with the highest predicted quality score) matches the candidate that is most similar to the human AD, measured as accuracy. I will also compare this selection accuracy to baselines such as always choosing the shortest candidate or always choosing the candidate generated with a specific prompt. If the selector improves this accuracy by at least 10–15% over these baselines, that would suggest that context-aware features are meaningfully improving AD selection.

Second, I will conduct a small user-feedback study with blind and low vision participants to assess whether the selected descriptions are actually useful in practice. Using targeted recruitment and snowball sampling, I will invite one or more blind or low vision contacts and, if possible, friends they recommend to participate in a brief remote session (see Figure A2 in Appendix A). For a subset of about 30–50 scenes, I will present three candidate ADs and ask the participants to rank them by usefulness and provide brief qualitative comments about clarity or timing. Because the sample will be small, I will treat this as an exploratory evaluation rather than a formal hypothesis test. The main quantitative signal will be the proportion of scenes for which the selection model and the participants both pick the same candidate AD, and qualitative feedback will help identify patterns where the selection appears reasonable according to metrics but fails to align with user preferences.

Qualifications: My interests in accessibility and machine learning highly align with this project. Previously, I was a research assistant under the URAP program where I learned how to write audio preprocessing functions to prepare data for specific model requirements. In 2025, I was awarded the WCAS Summer Grant Award to study how different supervised learning objectives affect the performance of deep neural networks on audio classification tasks. My background working with pretrained neural networks and large amounts of audio data is very relevant as I will be working with audio accessibility and multiple types of neural networks (MLLM, VLM, ASR). My STAT 303 coursework has taught me about regression and random forests, which are directly relevant to training my selector model. In the future, I hope to apply my machine learning skills to fields related to online accessibility or environmental science.

References

- Fang, B., Wu, W., Wu, Q., Song, Y., & Chan, A. B. (2025, March 8). *DistinctAD: Distinctive Audio Description Generation in Contexts*. Thecvf.com. <https://cvpr.thecvf.com/virtual/2025/poster/33034>
- Gao, Y., Fischer, L., Lintner, A., & Ebling, S. (2024). *Audio Description Generation in the Era of LLMs and VLMs: A Review of Transferable Generative AI Technologies*. ArXiv.org. <https://arxiv.org/abs/2410.08860>
- Kala, D., Khandelwal, E., & Tapaswi, M. (2025). *What You See is What You Ask: Evaluating Audio Descriptions*. ArXiv.org. <https://arxiv.org/abs/2510.00808>
- Li, C., Padmanabhuni, S., Cheema, M., Seifi, H., & Fazli, P. (2025). *VideoA11y: Method and Dataset for Accessible Video Description*. ArXiv.org. <https://arxiv.org/abs/2502.20480v1>
- Oncescu, A.-M., Henriques, J. F., Liu, Y., Zisserman, A., & Albanie, S. (2020). *QuerYD: A video dataset with high-quality text and audio narrations*. ArXiv.org. <https://arxiv.org/abs/2011.11071>
- Pitcher-Cooper, C., Seth, M., Kao, B., Coughlan, J. M., & Yoon, I. (2024). You Described, We Archived: A Rich Audio Description Dataset. *Journal on Technology and Persons with Disabilities : ... Annual International Technology and Persons with Disabilities Conference*, 11, 192. <https://pmc.ncbi.nlm.nih.gov/articles/PMC10956524/>
- World Wide Web Consortium (W3C). (n.d.). *How to meet WCAG (Quickref reference)*. Web Accessibility Initiative. <https://www.w3.org/WAI/WCAG22/quickref/?versions=2.1>
- Yang, A., Nagrani, A., Seo, P. H., Miech, A., Pont-Tuset, J., Laptev, I., Sivic, J., & Schmid, C. (2023). *Vid2Seq: Large-Scale Pretraining of a Visual Language Model for Dense Video Captioning*. ArXiv.org. <https://arxiv.org/abs/2302.14115>
- Ye, X., Chen, J., Li, X., Xin, H., Li, C., Zhou, S., & Bu, J. (2024). *CC BY-NC 4.0 MMAD: Multi-modal Movie Audio Description* (p. 11415). <https://aclanthology.org/2024.lrec-main.998.pdf>

Appendix A. Mentoring Plan and Consent

A1. Mentor Endorsement

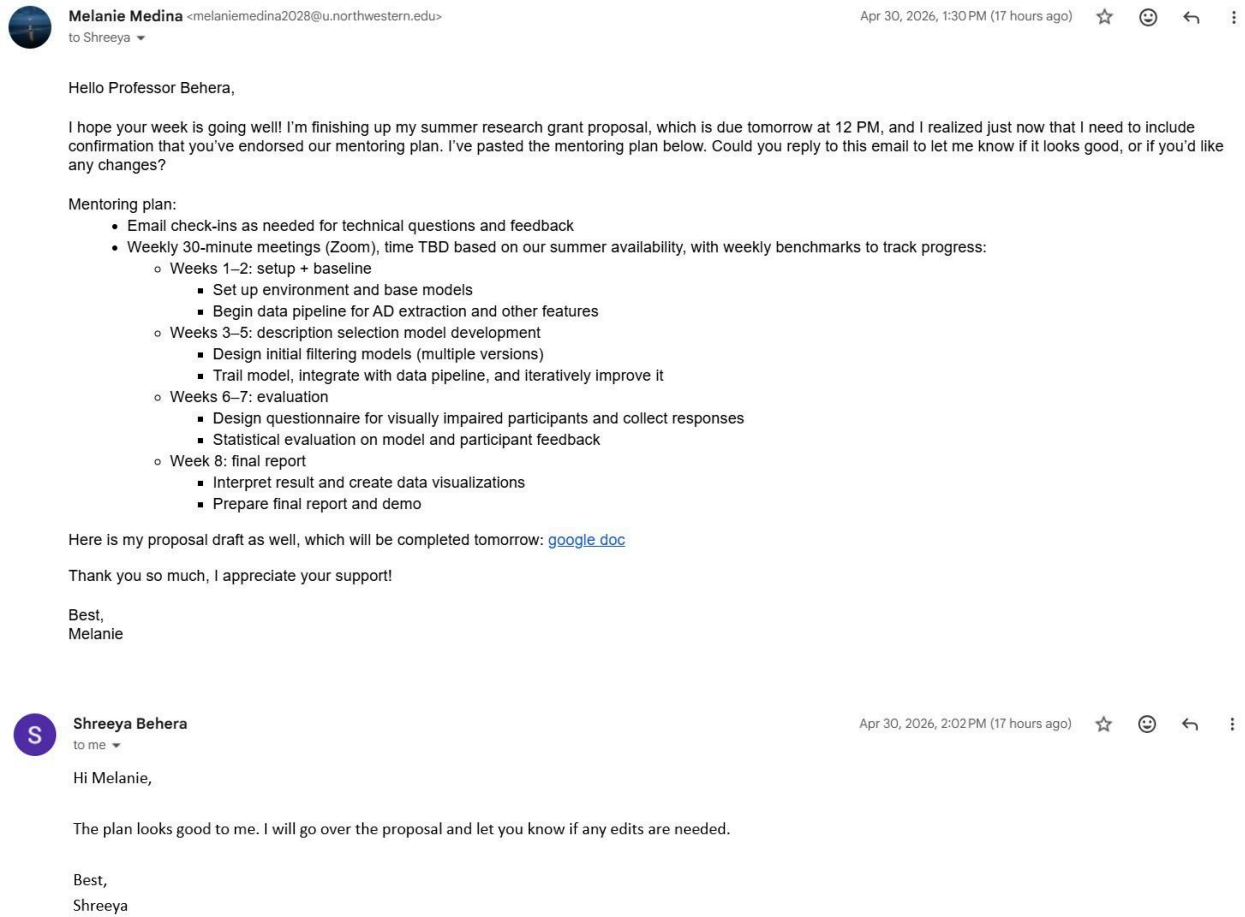
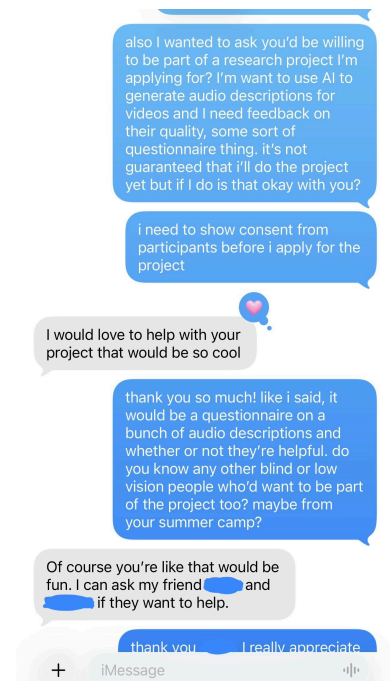


Figure A1. Screenshot of mentor's confirmation of the mentoring plan for this project.

A2. Blind Participant Agreement

Figure A2. Screenshot of informal conversation in which a blind participant expressed willingness to provide feedback on audio description samples.



Appendix B. Example Feature Table

B1. Sample Dataframe

scene_id	cand_id	scene_dur	speech_dur	silence_dur	n_silence_windows	n_visual_events	n_tokens	TTS_speech_dur	fits_silence	human_AD_sim	cov_score	redun_score	quality_score
s1	A	8.0	5.0	3.0	2	3	22	8.8	0	0.62	4	2	0.70
s1	B	8.0	5.0	3.0	2	3	15	6.0	1	0.75	5	1	0.88
s1	C	8.0	5.0	3.0	2	3	30	12.0	0	0.55	3	3	0.45
s2	A	5.5	4.5	1.0	1	1	10	4.0	1	0.60	4	2	0.68
s2	B	5.5	4.5	1.0	1	1	18	7.2	0	0.72	5	3	0.65
s2	C	5.5	4.5	1.0	1	1	9	3.6	1	0.58	3	1	0.62
s3	A	12.0	7.0	5.0	3	4	25	10.0	1	0.70	5	2	0.82
s3	B	12.0	7.0	5.0	3	4	14	5.6	1	0.65	4	1	0.73
s3	C	12.0	7.0	5.0	3	4	28	11.2	0	0.50	3	3	0.40

Table B1. Sample Dataframe from Phase A with 14 features and hypothetical numbers.

Column descriptions and how they are derived:

For training, **quality_score** is the response variable, and all other variables are predictors. At inference, the model will create a **predicted_score** for each candidate AD, which will be needed in evaluation.

scene_id: ID for the video segment from QuerYD/YuWA

cand_id: which prompt produced this candidate (A/B/C)

- A: simple prompt that asks for objective description of visuals
- B: longer detailed prompt that asks for description that follows AD guidelines
- C: prompt that will include both AD guidelines and dialogue transcript from ASR

scene_dur: scene duration in seconds, derived from QuerYD and YuWA provided timestamps

speech_dur: total seconds of speech in that scene from ASR

silence_dur: $\text{scene_dur} - \text{speech_dur}$

n_silence_windows: # of gaps in a scene that don't have dialogue/speech

n_visual_events: # of important visual events that happen in a scene, detected by VLM

n_tokens: token count of candidate AD text

TTS_speech_dur: estimated seconds of speech from candidate AD text-to-speech audio

- Using a simple equation like $\text{n_tokens} / 2.5$ words per sec

fits_silence: 1 if $\text{TTS_speech_dur} \leq \text{silence_dur}$, else 0

human_AD_sim: similarity between candidate AD and human AD, calculated using MLLM

cov_score: score from 1 (bad) to 5 (good) of how well the candidate AD covers info that is in the human AD

redun_score: score from 1 (not redundant) to 5 (super redundant) of how much the candidate AD repeats what is already obvious from dialogue

quality_score: score from 0 (bad) to 1 (good) of how good a candidate AD is relative to human AD and dialogue. Possible equation:

$$\text{quality_score} = 0.5 \cdot \text{human_AD_sim} + 0.3 \cdot \frac{\text{cov_score}}{5} + 0.2 \cdot \left(1 - \frac{\text{redun_score}}{5}\right)$$